STA2002 Tutorial 9

1. **Example: Testing normal** A manufacturing engineer is testing a power supply used in a notebook computer and, using $\alpha = 0.05$, wishes to determine whether output voltage is adequately described by a normal distribution. Sample estimates of the mean and standard deviation of $\bar{x} = 5.04$ V and $s = 0.08$ V are obtained from a random sample of $n = 100$ units.

   - A common practice in constructing the time intervals used in the chi-square test is to choose the intervals so that the expected frequencies p_i are equal for all cells.
   - Suppose we want to choose $k = 8$ cells, then the 8 intervals will be chosen such that $p_i = 1/8$

若 $X \sim N(\mu, \sigma^2)$

		probability p_i
A_1:	$(-\infty, \quad -z_{1/8}\sigma+\mu)$	0.125
A_2:	$[-z_{1/8}\sigma+\mu, -z_{1/4}\sigma+\mu)$	0.125
A_3:	$[-z_{1/4}\sigma+\mu, -z_{3/8}\sigma+\mu)$	0.125
A_4:	$[-z_{3/8}\sigma+\mu, \quad \mu)$	0.125
A_5:	$[\mu, \quad z_{3/8}\sigma+\mu)$	0.125
A_6:	$[z_{3/8}\sigma+\mu, z_{1/4}\sigma+\mu)$	0.125
A_7:	$[z_{1/4}\sigma+\mu, z_{1/8}\sigma+\mu)$	0.125
A_8:	$[z_{1/8}\sigma+\mu, +\infty)$	0.125

Class Interval	Observed Frequency o_i	Expected Frequency E_i
$x < 4.948$	12	12.5
$4.948 \le x < 4.986$	14	12.5
$4.986 \le x < 5.014$	12	12.5
$5.014 \le x < 5.040$	13	12.5
$5.040 \le x < 5.066$	12	12.5
$5.066 \le x < 5.094$	11	12.5
$5.094 \le x < 5.132$	12	12.5
$5.132 \le x$	14	12.5
Totals	100	100

H_0 : The form of the distribution is normal.
H_1 : The form of the distribution is nonnormal.

Test statistic: The test statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \frac{(12-12.5)^2}{12.5} + \frac{(14-12.5)^2}{12.5} + \cdots + \frac{(14-12.5)^2}{12.5} = 0.64$$

Since two parameters in the normal distribution have been estimated, the chi-square statistic above will have $k - d - 1 = 8 - 2 - 1 = 5$ degrees of freedom. We will use a fixed significance level test with $\alpha = 0.05$. Therefore, we will reject H_0 If $\chi^2 > \chi^2_{0.05}(5)$.

Conclusion: Since $\chi^2 < \chi^2_{0.05}(5) = 11.07$, fail to reject H_0 p-value: $\mathrm{P}\left(\chi^2(5) > 0.64\right) = 0.9861$

2. In 1992 poll 189 Montana residents were asked whether their personal financial status was worse, the same or better than one year ago. The opinions were divided into three groups by income range: under 20 K, between 20 K and 35 K, and over 35 K. We would like to test if opinions were independent of income.

	$b=3$			
$a=3$	Worse	Same	Better	
≤ 20 K	20	15	12	47
(20 K, 35 K)	24	27	32	83
≥ 35 K	14	22	23	59
	58	64	67	189

The chi-squared statistic is

$$T = \frac{(20 - 47 \times 58/189)^2}{47 \times 58/189} + \ldots + \frac{(23 - 67 \times 59/189)^2}{67 \times 59/189} = 5.21.$$

If we take level of significance $\alpha = 0.05$, then the threshold is:

$$\chi^2_{(a-1)(b-1),0.05} = \chi^2_{4,0.05} = 9.488.$$

Since $T = 5.21 < 9.488$, we accept the null hypothesis that opinions are independent of income.

3. In a study of the television viewing habits of children, a developmental psychologist selects a random sample of 300 first graders - 100 boys and 200 girls. Each child is asked which of the following TV programs they like best: The Lone Ranger, Sesame Street, or The Simpsons. Results are shown in the contingency table below. The question is, Do the boys' preferences for these TV programs differ significantly from the girls' preferences? Use a 0.05 level of significance.

	Lone Ranger	Sesame Street	The Simpsons	Total
Boys	50	30	20	100
Girls	50	80	70	200
Total	100	110	90	300

$$\begin{aligned}
&\mathrm{df} = (\mathrm{r}-1)^*(\mathrm{c}-1) = (2-1)^*(3-1) = 2\\
&\mathrm{E}_{\mathrm{r,c}} = (\mathrm{n_r}^*\mathrm{n_c})\,/\mathrm{n}\\
&\mathrm{E}_{1,1} = (100*100)/300 = 10000/300 = 33.3\\
&\mathrm{E}_{1,2} = (100*110)/300 = 11000/300 = 36.7\\
&\mathrm{E}_{1,3} = (100*90)/300 = 9000/300 = 30.0\\
&\mathrm{E}_{2,1} = (200*100)/300 = 20000/300 = 66.7\\
&\mathrm{E}_{2,2} = (200*110)/300 = 22000/300 = 73.3\\
&\mathrm{E}_{2,3} = (200*90)/300 = 18000/300 = 60.0
\end{aligned}$$

$$\begin{aligned}
\mathrm{X}^2 &= (50-33.3)^2/33.3 + (30-36.7)^2/36.7 + (20-30)^2/30 + (50-66.7)^2/66.7 + (80-73.3)^2/73.3 + (70-60)^2/60\\
&= 8.38 + 1.22 + 3.33 + 4.18 + 0.61 + 1.67\\
&= 19.39
\end{aligned}$$

The p-value is $P\left(X^2 > 19.39\right) = 0.00006$. Hence, we conclude that the boys' preferences for these TV programs differ significantly from the girls' preferences.

4. The head of a surgery department at a university medical center was concerned that surgical residents in training applied unnecessary blood transfusions at a different rate than the more experienced attending physicians. Therefore, he ordered a study of the 49 Attending Physicians and 71 Residents in Training with privileges at the hospital. For each of the 120 surgeons, the number of blood transfusions prescribed unnecessarily in a one-year period was recorded. Based on the number recorded, a surgeon was identified as either prescribing unnecessary blood transfusions Frequently, Occasionally, Rarely, or Never. Here's a summary table (or "contingency table") of the resulting data:

Physician	Frequent	Occasionally	Rarely	Never	Total
Attending	2(4.1%)	3(6.1%)	31(63.3%)	13(26.5%)	49
Resident	15(21.1%)	28(39.4%)	23(32.4%)	5(7.0%)	71
Total	17	31	54	18	120

Are attending physicians and residents in training distributed equally among the various unnecessary blood transfusion categories?
The observed data were given to us in the table above. So, the next thing we need to do is find the expected counts for each cell of the table:

Physician	Frequent	Occasionally	Rarely	Never	Total
Attending					49
Resident					71
Total	17	31	54	18	120

It is in the calculation of the expected values that you can readily see why we have $(2-1)(4-1)=3$ degrees of freedom in this case. That's because, we only have to calculate three of the cells directly.
Recall that

$$E(n_{i,j}) = \frac{\text{row sum x colum sum}}{\text{grand total}},$$

we have:

$$E(n_{1,1}) = \frac{17*49}{120} = 6.942, E(n_{1,2}) = \frac{31*49}{120} = 12.658, E(n_{1,3}) = \frac{54*49}{120} = 22.05$$

Physician	Frequent	Occasionally	Rarely	Never	Total
Attending	6.942	12.658	22.05		49
Resident					71
Total	17	31	54	18	120

Once we do that, the remaining five cells can be calculated by way of subtraction.
For example, $E(n_{2,1}) = 17 - 6.942 = 10.058$

Physician	Frequent	Occasionally	Rarely	Never	Total
Attending	6.942	12.658	22.05	7.35	49
Resident	10.058	18.342	31.95	10.65	71
Total	17	31	54	18	120

Alternatively, you can use the above formula to get the values.

Now that we have the observed and expected counts, calculating the chi-square statistic is straightforward:

$$Q = \frac{(2-6.942)^2}{6.042} + \ldots + \frac{(5-10.65)^2}{10.65} = 31.88$$

Because $\chi^2_{3,0.05} = 7.815$, we reject the null hypothesis at the 0.05 level. There is sufficient evidence at the 0.05 level to conclude that the distribution of unnecessary transfusions differs among attending physicians and residents.